



United International University
QUEST FOR EXCELLENCE



CSE 1326: Digital Logic Design Lab

Multiplexers

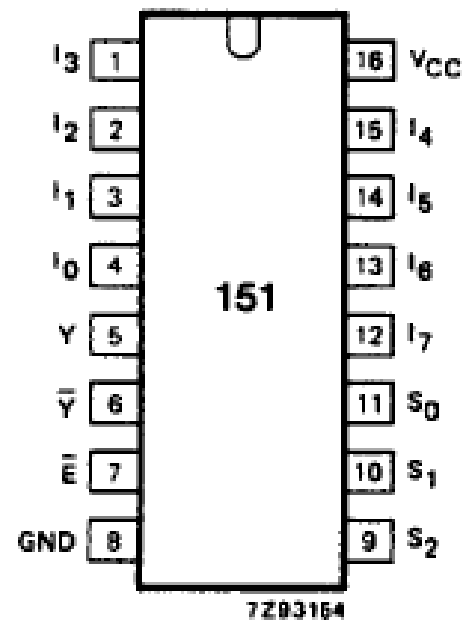
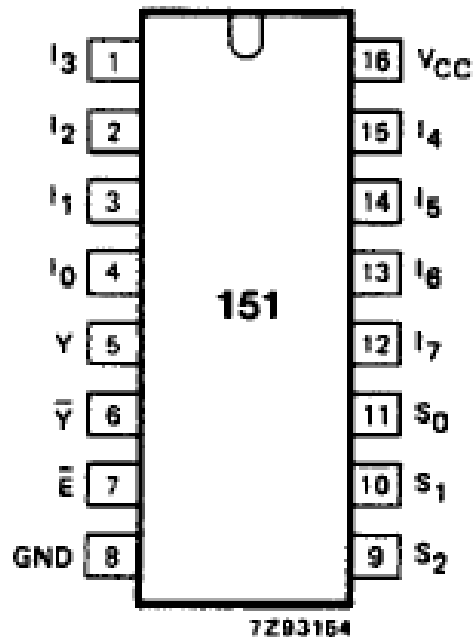
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Objective

- Study 4-input and 8-input multiplexers
- Design and implement the Boolean Function $f(A,B,C,D) = \sum_m (0,1, 2, 4, 7, 9, 14,15)$ using
 1. 8 input MUX and necessary basic gates
 2. 4 input MUX and necessary basic gates

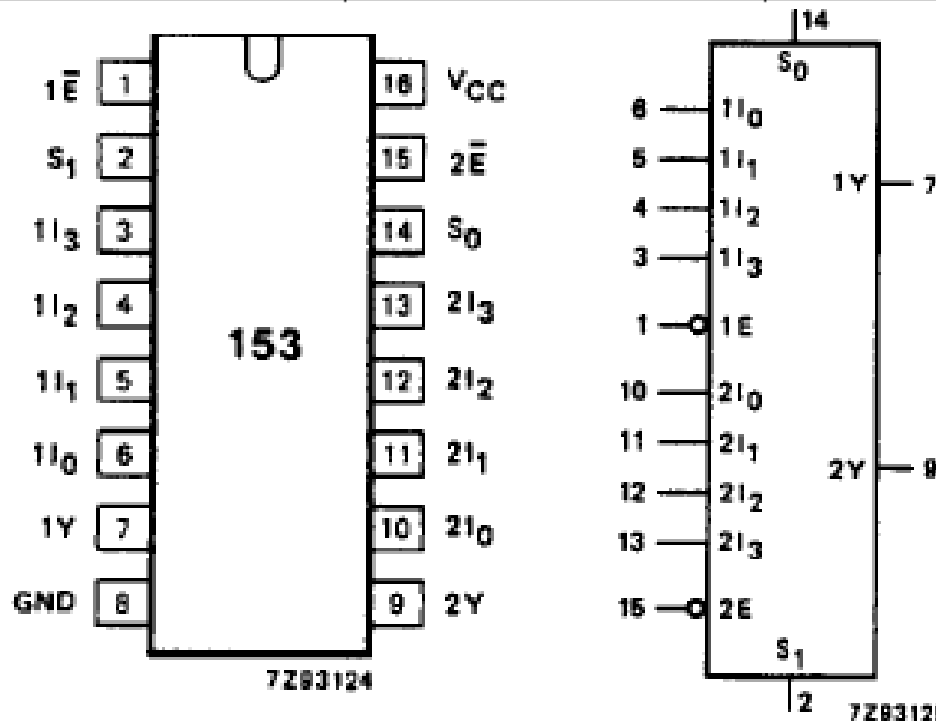
74151: 8-input multiplexer

PIN NO.	SYMBOL	NAME AND FUNCTION
4, 3, 2, 1, 15, 14, 13, 12	I_0 to I_7	multiplexer inputs
5	Y	multiplexer output
6	$\bar{Y}$	complementary multiplexer output
7	$\bar{E}$	enable input (active LOW)
8	GND	ground (0 V)
11, 10, 9	S_0, S_1, S_2	select inputs
16	V_{CC}	positive supply voltage



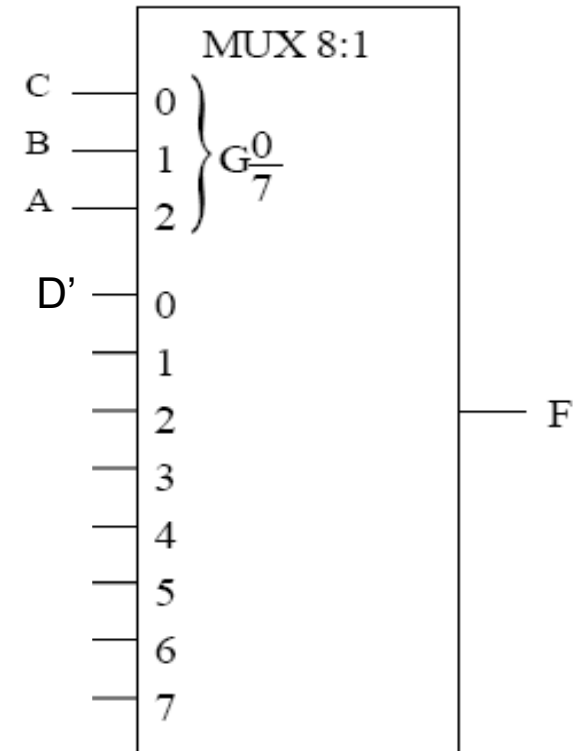
74153: Dual 4-input multiplexer

PIN NO.	SYMBOL	NAME AND FUNCTION
1, 15	$1\bar{E}, 2\bar{E}$	output enable inputs (active LOW)
14, 2	S_0, S_1	common data select inputs
6, 5, 4, 3	$1I_0$ to $1I_3$	data inputs from source 1
7	$1Y$	multiplexer output from source 1
8	GND	ground (0 V)
9	$2Y$	multiplexer output from source 2
10, 11, 12, 13	$2I_0$ to $2I_3$	data inputs from source 2
16	V_{CC}	positive supply voltage



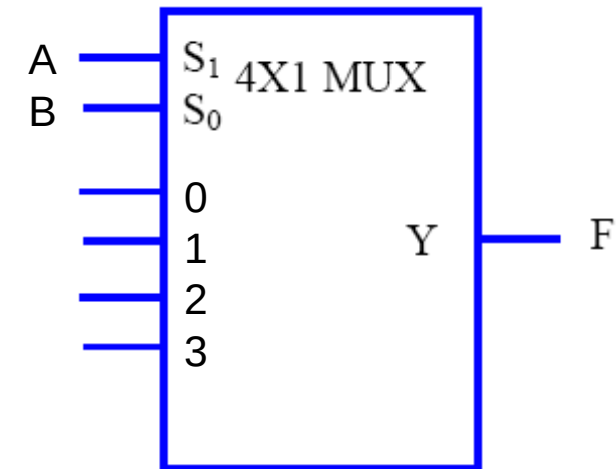
For 8-input multiplexer

A	B	C	D	F
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1



For 4-input multiplexer

A	B	C	D	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1



Writing Report

- Description of 74151 and 74153 (truth table, pin diagram)
- Detailed description of both implementation $f(A,B,C,D) = \sum_m (0,1, 2, 4, 7, 9, 14,15)$ using
 1. 8 input MUX and necessary basic gates
 2. 4 input MUX and necessary basic gates
- Description of design for the Boolean Function $f(A,B,C,D) = \sum_m (0,1, 2, 4, 7, 9, 14,15)$ using 2-input MUX and necessary basic gates